DIPLOMA PROGRAMME PRACTICE · SL · SET B

Chemistry / Paper 1B

Question booklet · 25 marks · 90 minutes across 1A and 1B

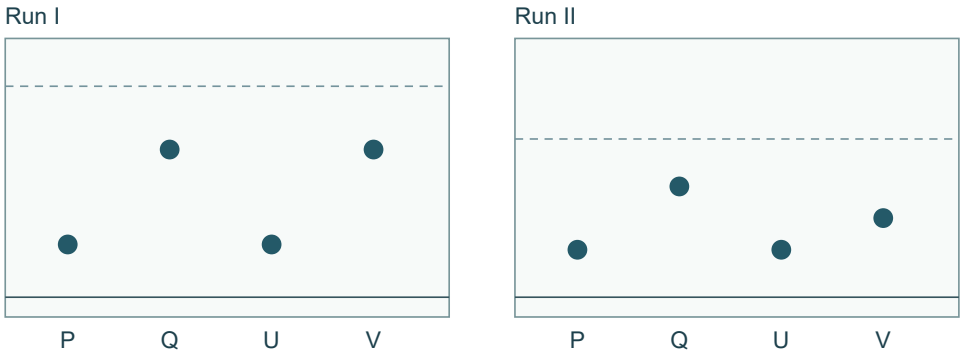
Answer every part. Show working, units and suitable precision. Use a calculator and a clean Chemistry data booklet. Use separate answer paper for additional working and diagrams.

QUESTION 1 [11]

Two solvents for a plant extract

The same extract and pure standards P and Q are run on polar silica with two different mobile phases. Distances are measured from the baseline. Rf is spot distance divided by solvent-front distance. Spots U and V belong to the extract; each lane is developed under the stated common conditions.

Run	Front / cm	P / cm	Q / cm	U / cm	V / cm
I	8.0	2.0	5.6	2.0	5.6
II	6.0	1.8	4.2	1.8	3.0



TLC plates for runs I and II. Solid line: baseline; dashed line: solvent front. U and V are shown in separate lanes for comparison.

(a) Calculate Rf values for V and Q in each run. **[3]**

(b) Assess the identifications U=P and V=Q, using both runs. **[3]**

(c) Within run I, compare how the phases affect P and Q. Explain why absolute migration distances alone cannot be compared between the two runs. **[3]**

Paper 1B · Question 1 (continued)

(d) Predict the effect of submerging the starting spots. State one further practical precaution for obtaining reliable Rf values. **[2]**

QUESTION 2 [14]

Standardizing a monoprotic acid

A student dissolves 1.325 g of dry Na_2CO_3 (M=106.0 g mol^{-1}) and makes the solution to 250.0 cm^3. A 25.00 cm^3 aliquot is titrated with HCl to complete reaction, forming CO_2 and water. Each burette reading has absolute uncertainty ±0.05 cm^3.

Trial	Initial / cm^3	Final / cm^3
Rough	0.00	25.40
1	0.10	25.00
2	0.30	25.25
3	0.15	25.05

(a) Determine the three accurate titres and their mean. Explain the treatment of the rough trial. **[3]**

(b) Calculate the concentration of the sodium carbonate solution and the amount in each aliquot. **[3]**

(c) Write the ionic equation and determine the HCl concentration. **[3]**

(d) Estimate the maximum percentage uncertainty from the two burette readings for a single accurate titre. Explain why close titres do not rule out a calibration error. **[3]**

(e) Suppose the carbonate had absorbed water before weighing. Deduce the direction of error in the calculated acid concentration if the full mass is treated as anhydrous. **[2]**